

TABLE OF CONTENTS

M-422 CENTRAL LABORATORY

	Page No.
1. SCOPE OF WORK	M-422-2
2. CODES AND STANDARDS	M-422-2
3. DETAIL REQUIREMENTS	M-422-3
4. REAGENT LIST	M-422-10
5. DESIGN CRITERIA.....	M-422-17
6. DETAILED REQUIREMENTS	M-422-18
7. PERFORMANCE TESTING AND GUARANTEES	M-422-19
8. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC).....	M-422-20
9. INSTRUMENTATION AND CONTROLS	M-422-20
10. ELECTRICAL SPECIFICATION	M-422-20

M-422 CENTRAL LABORATORY

1. SCOPE OF WORK

Bidder shall perform the work included in this specification in accordance with Part 4, Subsection 1.2, “Summary of Work”.

The work shall include the buildings, laboratory equipment and chemicals to conduct all the specified analysis and tests until the end of guarantee Period of the Muara Tawar CCPP (Block 2, 3 & 4) Add-On Project Power Plant.

The Bidder shall provide one (1) set of complete Central Laboratory equipment for block 2, 3 & 4 requirements.

The Bidder shall furnish central Laboratory complete with all necessary furniture such as corrosion resistant tables with sink and electrical plugs, smoke hoods, cupboards and experimental desk including all the reagent detailed on clause 4 herein and the equipment detailed in tables 1 to 2.

The Bidder shall propose the proper and suitable layout to keep all the equipment listed in table 1 to 2 and shall provide special desk to conduct special Laboratory analysis.

All laboratory equipment shall be of the highest quality and suitable for use over the plant life.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

The codes and standards from Part 4, Subsection 2.5 are specifically applicable to the Work involved in this specification.

AGMA	American Gear Manufacturer Association
ASTM	American Society for testing and Material
CEMA	Conveyer Equipment Manufacturer's Association
IEC	International Electrotechnical Commission
ISO	The International Organization for Standardization
SPLN	Standards PLN

Further to the standard test specified in Table 1 through 2 per specified codes and standards that the laboratory shall be capable of conducting.

3. DETAIL REQUIREMENTS

Table 1: GENERAL LABORATORY APPARATUS

No.	Apparatus Name:	Specification	Amount
1.	Measuring Apparatus : Analyst Weigher	Single bowl, direct reading Capacity : 200 g Range : 0- 200 g Accuracy : 0,5	1 pc
2.	Precision Weigher	Direct Reading Capacity : 1000 g Range : 0- 1000 g Accuracy : 0.05 g	1 pc
3.	Dead Weigher	Direct Reading : Capacity : 150 kg Range : 0 – 150 Smallest Scale : 0.05 kg	1 pc
4.	Pocket Thermometer	Max temperature : 150°C Range : 30 °C + 150°C Smallest Scale : 1°C	1 pc
5.	Interval timer	Electric with sound alarm Capacity : 60 minutes Smallest Scale : 1 minute	1 pc

No.	Apparatus Name:	Specification	Amount
6.	Stop watch	One turn : 60 seconds Smallest Scale : 1/5 second Registered : 30 minutes with one second scale	2 pcs
7.	Stop watch	One turn : 30 seconds Smallest Scale : 1/10 second Registered : 30 minutes with 1/5 second scale	1 pc
	Glass Apparatus :		
8.	Beaker Glass	Pyrex, short with scale approximate scale Sizes : 100 ml : 250 ml : 400 ml : 600 ml : 1000 ml	20 pcs 20 pcs 20 pcs 20 pcs 5 pcs
9.	Beaker Glass	Pyrex, tall with approximate scale Size : 300 ml : 1000 ml	10 pcs 10 pcs
10	Reagent Bottle	Pyrex, labeled with approximate scale 500 ml Clear colour Dark colour Pyrex, labeled with cover 250 ml Clear colour Dark colour	10 pcs 10 pcs 10 pcs 10 pcs
11.	Burette	Class A with cover made of teflon Sizes : 10 ml, scale 0.05 ml : 50 ml, scale 0.1 ml	2 pcs 2 pcs
12	Burette	Capacity 50 ml, bottle 2000 ml, scale 0,1 ml Capacity 10 ml, bottle 1000 ml, scale 0,05 ml	2 pcs 2 pcs
13	Measuring Glass	Size : 50 ml : 100 ml : 500 ml	5 pcs 5 pcs 5 pcs
14	Measuring Glass	Pyrex Sizes : 1000 ml	2 pcs
15	Desicator	Pyrex, diameter in 250 mm, complete with desicator	1 pc
16	Evaporator disk	Porcelain Size : 250 ml	1 pc
17	Watch Glass	Pyrex, diameter : 100 mm	6 pcs

No.	Apparatus Name:	Specification	Amount
18	Erlen Mayer Flask	Narrow mouth Size : 50 ml : 250 ml : 1000 ml	10 pcs 10 pcs 3 pcs
19	Filtering Flask	With nipple for tube connection : 200 ml : 500 ml	2 pcs 2 pcs
20	Measuring Flask	Class A, with cover Size : 100 m : 500 m	4 pcs 4 pcs
21	Glass Tubing	Pyrex, standard, diameter Length : 5 ml : 10 ml	5 pcs 5 pcs
22	Glass Rod	Pyrex, standard, out side diameter : 3 mm : 5 mm : 8 mm	5 pcs 5 pcs 5 pcs
23	Nessler Tube	Height Size : 50 ml : 100 ml	5 pcs 5 pcs
24	Scalled Pipette	Class A Size : 1 ml : 2 ml : 5 ml : 10 ml : 25 ml : 50 ml : 100 ml	5 pcs 5 pcs 5 pcs 5 pcs 5 pcs 8 pcs 4 pcs
25	Mohr pipette	Class A Size : 1 ml : 5 ml : 10 ml : 50 ml	5 pcs 5 pcs 5 pcs 5 pcs

No.	Apparatus Name:	Specification	Amount
26	Portable demineralizer	Capacity : 120 l/hour	1 set
27	Other Heater	Multipurpose, multimantle	1 pc
28	Stirrer	Rust proof steel	1 pc
29	Magnetic Stirrer	110 x 110 (mm)	1 pc
30	Pinset	General usage with teflon	
		Length : 120 cm	1 pcs
		: 100 cm	1 pcs
31	Funnel	Short neck kimer, 100 mm	5 pcs
		To pour, powder, with handle	5 pcs
32	Gloves	Rubber	2 pcs
		Asbestos	2 pcs
33	Mask	Can cover the whole face	2 pcs
34	Safety glass	Plastic with sides cover	2 pcs
35	Mortar with pestle	Agate	
		Size : 60 mm	1 pc
36	Magnet rode	Size : 66 mm x 20 mm x 150 mm	1 pc
37	Toolkit	For laboratory equipments	1 set
38	Vacuum pump	Capacity	
		Vacuum : 25 l/min	1 set
		: 650mm Hg	
39	Pipette filler	Rubber	2 pcs
40	Rubber bulb	Pressurised or vacuum, both ways	2 pcs
41	Support grid		1 set

No.	Apparatus Name:	Specification	Amount
42	Funnel support		1 pc
43	Pipette support	Plastic	1 set
44	Test tube rack	Wood on plastic	2 pcs
45	Beaker clamp		1 pc
46	Tubes	Tygon Size : 6,4 mm x 3,2 mm : 9,5 mm x 3,2 mm : 13 mm x 3,2 mm	20 mtr 20 mtr 20 mtr
47	Cork borer	Size : 5 mm - 22 mm	1 set
48	Cork borer Sharpener	Size : 22 mm	1 unit
49	Bottle lids	Various size : 0 - 8 - From cork - From rubber	1 pack (180)pcs 1 pack (180)pcs
50	Clamp	Small, made from plastic Small, made from asbest	2 unit 2 unit
51	Clamp holder	Castaloy can be used for rods up to diameter of 19 mm	10 unit
52	Small extension Clamp	Castaloy R, round clamp vinyl coated length 8 inch clamp opening 1 m - 1 1/2 m	2 unit
53	Ring extension	Concentric, iron or with integrad tank Size : 7,6 mm : 15,2 mm	2 unit 2 unit
54	Ring support	Concentric, iron with integral clamp and copper screw Size : 7,6 cm (3inch) : 15,2cm(6inch)	1 Pc 1 Pc

No.	Apparatus Name:	Specification	Amount
55	Treelegs	Iron with concentric ring	2 pcs
56	Nessler Tube support		2 pcs
57	Bottle	Plastic Capacity : 500 ml : 1000 ml	10 pcs 15 pcs
58	Bottle	Plastic with pipette 60 ml	10 pcs
59	Bottle	Wide mouth plastic Capacity : 500 ml : 250 ml	10 pcs 10 pcs
60	Wash bottle	Plastic Capacity : 500 ml	3 pcs
62	Boiling chip	Can not evaporate	1 bottle
63	Flask brush	From hard ware Size : 35 cm (15 inch) : 56 cm (22 inch)	
64	Tube brush	Size : 65 ml	2 unit
65	Burette clamp		3 pcs
66	Refrigerator	Size : 170 l (6 ft ³)	1 unit

SPLN 19:1979

Table II: APPARATUS FOR WATER ANALYSIS

No	Apparatus Name	Specification	Analysis	Method	Amount	Category
1	Conductivity meter	Cell type : dip cell Automatic temperature compensator Capacity : 0 — 20 000 $\mu\text{u/cm}$ Range selector resistance 0 - 2 $\mu\text{u/cm}$ 0 - 20 $\mu\text{u/cm}$ 0 - 200 $\mu\text{u/cm}$ 0 - 2000 $\mu\text{u/cm}$ 0 - 20000 $\mu\text{u/cm}$ Accuracy $\pm 2\%$ full scale Power Battery and AC 110/220 volt 50 Hz	Water Conductivity	ASTM D 1125	1 set	A
2	pH - meter	Range : 0 - 14 pH Scale : 0,1 pH Accuracy: 0,05 pH Temperature Compensator 0 - 100°C	pH water	ASTM D 1293	1 set	A
3	Spectrofotometer	Double beam Range of wave length : 190 — 900 mm Accuracy: 0,005 A Power : AC 110/220 volts, 50 Hz	Ions cone - Iron - Phosphate - Chlor - Ammonium - Silica - Hydrazine - Copper - Sulphite - Chlorine - Dissolved oxygen	ASTM D 516 D 515 D 512 D 1426 D 859 D 1385 D 1688 D 1339 D 888 D 1291		

No	Apparatus Name	Specification	Analysis	Method	Amount	Category
4	Color Comparator	To find Cl ₂ (Chlorine) Range : 0 - 1,0 ppm scale 0,1 1,0 - 5,0 ppm	Chlorine	-	1 set	A
		Scale 0,5 To find DO ₂ (Dissolved oxygen) Range : 0 - 50ppb	Dissolved oxygen	-	1 set	A
5	Jackson can le Turbidity meter	Range :	Water turbidity	ASTM : D 1889	1 set	C

4. REAGENT LIST

The grade of all reagents shall be of superior quantity.

	<u>Material</u>	<u>Specifications</u>	<u>Chemical Formula</u>	<u>Quantity</u>
(1)	Sulphuric acid	Concentrated	H ₂ SO ₄	5 l
(2)	Phenolphthalein	Indicator	-	50 gr
(3)	Ethyl alcohol		C ₂ H ₅ OH	2 l
(4)	Methyl orange	Indicator	-	50 gr
(5)	Hydrochloric acid	Concentrated	HCl	5 l
(6)	Strontium chlorid		SrCl ₂ 6H ₂ O	Smallest Package
(7)	Barium chloride		BaCl ₂ 2H ₂ O	500 gr
(8)	Sodium Hydroxide		NaOH	2 kg
(9)	Potassium chromate		K ₂ CrO ₄	1 kg
(10)	Silver nitrate		AgNO ₃	1 kg
(11)	Sodium chloride		NaCl	2 kg
(12)	Mercuric nitrate		Hg(NO ₃) ₂ 2H ₂ O	200 gr

	<u>Material</u>	<u>Specifications</u>	<u>Chemical Formula</u>	<u>Quantity</u>
(13)	Nitric acid	Concentrated	HNO ₃	3 l
(14)	Dyphenyl carb	Indicator	-	Smallest Package
(15)	Bromophenol blue	Indicator	-	Smallest Package
(16)	Ammonium oxalate		(NH ₄) ₂ C ₂ O ₄ .H ₂ O	500 gr
(17)	Ammonium hydroxide	Concentrated	NH ₄ OH	2 l
(18)	Potassium permanganate	0.1N	KMnO ₄	5 ea.
(19)	Sodiumthiosulphate	0.1N	Na ₂ S ₂ O ₃ .5H ₂ O	5 ea.
(20)	Sodiumthiosulphate		Na ₂ S ₂ O ₃ .5H ₂ O	500 gr
(21)	Starch		(C ₆ H ₅) ₅ n	500 gr
(22)	Sodium sulphide		Na ₂ S.9H ₂ O	500 gr
(23)	Potassium sodiumtartrate		KNaC ₂ H ₄ O ₆ .4H ₂ O	500 gr
(24)	Disodium EDTA		-	1 kg
(25)	Magnesium chloride		MgCl ₂ .6H ₂ O	500 gr
(26)	Ammonium chloride		NH ₄ Cl	500 gr
(27)	Eriochrome black T	Indicator	-	100 gr
(28)	Sodium Carbonate		Na ₂ CO ₃	500 gr
(29)	Methyl alcohol		CH ₃ OH	1 l
(30)	Calcium .carbonate		CaCO ₃	500 gr
(31)	Methyl red	Indicator	-	25 kg
(32)	P-Dimethyl amino benzaldehyde		4-(CH ₃) ₂ NC ₆ H ₄ CHO	100 gr
(33)	Hydrazine dichloride		N ₂ H ₄ . HCl	500 gr
(34)	Glycerin		-C ₃ H ₅ -(OH) ₃	1 l
(35)	Potassium iodide		KI	1 kg

	<u>Material</u>	<u>Specifications</u>	<u>Chemical Formula</u>	<u>Quantity</u>
(36)	Indigo carmen		-	100 gr
(37)	Potassium hydroxide		KOH	2 kg
(38)	Dextrose		-C ₆ H ₁₂ O ₆	500 gr
(39)	Cobalt chloride hexa hydrate		CaCl ₂ .6H ₂ O	500 gr
(40)	Ferric III Chloride		FeCl ₃ .6H ₂ O	500 gr
(41)	Copper sulphate		CuCO ₄ .5H ₂ O	500 gr
(42)	Sodium silicate		Na ₂ SiO ₃ .9H ₂ O	500gr
(43)	Buffer tablet for hardness determina - tion with EDT			2 bottles each containing 1000 tablets
(44)	Ammonium acetate		NH ₄ C ₂ H ₃ O ₂	1 kg
(45)	Acetik acid glasil		CH ₃ COOH	2 l
(46)	Kongo red Indicator		-	100 gr
(47)	Chloroform		-	1 l
(48)	Cupper (metal)		-	10 gr
(49)	Sodium citrate		C ₃ H ₄ (OH)(CO ₂ Na) ₃ .2 H ₂ O	1 kg
(50)	Sodium acetate		NaC ₂ H ₃ O ₂ .3H ₂ O	1 kg
(51)	Amino naphthol Sulphuric acid		-	500 gr
(52)	Sodium sulfite		Na ₂ SO ₃	1 kg
(53)	Sodium pyrrosulfite		Na ₂ S ₂ O ₅	1 kg
(54)	Ammonium meta vanadate		NH ₄ VO ₃	500 gr
(55)	Potassium mono phosphate		KH ₂ PO ₄	500 gr
(56)	Tin chloride		SnCl ₂ .2H ₂ O	500 gr

	<u>Material</u>	<u>Specifications</u>	<u>Chemical Formula</u>	<u>Quantity</u>
(57)	Sodium chromate		NaCrO ₄	500 gr
(58)	Oxalic acid		H ₂ C ₂ O ₄	1 kg
(59)	Potassium hydrogen pothalate		KHC ₈ H ₄ O ₄	1 kg
(60)	Disodiumphosphate		Na ₂ HPO ₄	500 gr
(61)	Hydrofluoric acid		HF	500 gr
(62)	Perchloric acid		HClO ₄	1 ltr
(63)	Isopropil alcohol		C ₃ H ₈ O	2 ltr
(64)	N-Hexyl alcohol		CH ₃ (CH ₂) ₄ CH ₂	2 ltr
(65)	Iso-amylachol		-	2 ltr
(66)	Baso-phenontrolin		-	10 gr
(67)	Orthepeanthroline		C ₁₂ H ₈ N ₂ .HCl.H ₂ O	25 gr
(68)	Murexide inic. tablets containing			5 bottles each 100 tablets
(69)	Ammonium thiosyanaide		NH ₄ SCN	500 gr
(70)	Potassium chloride		-	1 kg
(71)	Bromocresol green	Indicator	-	smaller package
(72)	Thynol blue		N ₂₇ H ₃₀ O ₅ S	-
(73)	Triethanol		-	500 gr
(74)	Zinc (metal)		-	smallest package
(75)	Zinc chloride		-	500 gr
(76)	Potassium		-	1 kg
(77)	Hydroxylamine hydrochloride		NH ₂ OH.HCl	1 kg

	<u>Material</u>	<u>Specifications</u>	<u>Chemical Formula</u>	<u>Quantity</u>
(78)	Manganium sulphate		$\text{MnSO}_4 \cdot 4\text{H}_2\text{O}$	1 kg
(79)	Potassium iodate		KIO_3	1 kg
(80)	Sodium Hydrogencarbonate		NaHCO_3	500 gr
(81)	Sodium agide		NaN_3	smallest package
(82)	Barium chloride		$\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$	500 gr
(83)	Ammonium ferric sulphate		$\text{Fe}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4$	500 gr
(84)	Potassium thiocyanate		KSCN	500 gr
(85)	Bromathymal	Indicator blue		Smallest package
(86)	Calcium chloride		$\text{CaCl}_2 \cdot \text{H}_2\text{O}$	500 gr
(87)	Potassium nitrate		-	500 gr
(88)	Tartaric acid		$\text{C}_2\text{H}_2(\text{OH})_2(\text{COOH})_2$	1 kg
(89)	Mercuric chloride		HgCl_2	500 gr
(90)	Zinc sulphate		$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$	500 gr
(91)	Iodine, reblimed		I_2	500 gr
(92)	Basocaproin		$(\text{CH}_3)_2(\text{C}_6\text{HS})_2 \cdot \text{C}_{12}\text{H}_4\text{N}_2$	Smallest package
(93)	Zincon		$\text{C}_6\text{H}_3(\text{SO}_3\text{H})(\text{OH})\text{N}_2 \cdot 0\text{C}_6\text{H}_5$	smallest package
(94)	Hystenol orange	Indicator	-	
(95)	Ammonium ferrous sulphate		$\text{Fe SO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$	1 kg
(96)	Sulphosalicylic acid		-	smallest package
(97)	Ammonium persulphate		$(\text{NH}_4)_2\text{S}_2\text{O}_8$	500 gr
(98)	Ammonium citrate		$(\text{NH}_4)_2\text{BC}_6\text{H}_5\text{O}_7$	1 kg

<u>Material</u>	<u>Specifications</u>	<u>Chemical Formula</u>	<u>Quantity</u>
(99) Dimethylglyoxime .		$(CH_3)_2C_2(NO_2)_2$	500 gr
(100) Ammonium nickelsulphate		$NiSO_4(NH_4)_2SO_4 \cdot 6H_2O$	Smallest package
(101) Silversulphate	Indicator	$AgSO_4$	1 kg
(102) Sodium oxalate		$Na_2C_2O_4$	1 kg
(103) Sodium chloride		$NaOCl$	500 ml
(104) Aluminium sulphate		$Al_2(SO_4)_3 \cdot 18H_2O$	1 kg
(105) Normal hexane			2ltr
(106) Para-phenylenediamine		$NH_2 \cdot C_6H_4 \cdot NH_2$	smallest package
(107) Borax		$Na_2B_4O_7 \cdot 10H_2O$	500 gr
(108) Silver chloride		$AgCl$	500 gr
(109) Magnesium oxide		MgO	100 gr
(110) Fluorescein-Na		$C_{20}H_{10}O_5Na_5$	500 gr
(111) Sodium acetate (trihydrate)		$CH_3COONa \cdot 3H_2O$	500 gr
(112) Ammonium molybdate		$(NH_4)_2MoO_4$	500 gr
(113) Sodium hydrogensulphate		$NaHSO_3$	500 gr
(114) Phenol		C_6H_5OH	Smallest package
(115) Acetone		CH_3COCH_3	500 gr
(116) Cuprizone	Indicator	$C_6H_{10}NNHCOCONH-NC_6H_{10}$	Smallest package
(117) Triethanolamine		$N(CH_2CH_2OH)_3$	500gr
(118) D.D.C	Indicator	$(C_2H_5)_2NCS_2Na \cdot 3H_2O$	Smallest package
(119) T.P.T.Z	Indicator	$(C_5H_4N)_3C_3N_3$	Smallest package

	<u>Material</u>	<u>Specifications</u>	<u>Chemical Formula</u>	<u>Quantity</u>
(120)	Nitrobenzene		$C_6H_5NO_2$	Smallest package
(121)	Sodium perchlorate		$NaClO_4$	500 gr
(122)	Sulphanilic acid		$NH_2C_6H_4SO_2NH_2$	500 gr
(123)	3,3-Dimethyl-Indicator benzidine		$Cl_4H_{16}N_2$	Smallest package
(124)	Potassium dichromate		$K_2Cr_2O_7$	500 gr
(125)	Sodium phrophosphate		$Na_4P_2O_7 \cdot 10H_2O$	500 gr
(126)	Formaldehyde		$HCHO$	500 gr
(127)	Brucine Indicator dyhydrochloride		$C_{23}H_{26}N_2O_4$	100 gr
(128)	-N(1-Naphthyl) ethylenediamine	Indicator	$ClOH_7NHCH_2CH_2NH_2 \cdot 2HCl$	Smallest
(129)	-N,N-Dimethylp-phenylenediamine package	Indicator	$C_6H_4(NH_2)(N(CH_3)_2)$	Smallest
(130)	Calcium standard Solution		Ca	Smallest Package
(131)	Cadmium standard solution		Cd	smallest package
(132)	Copper standard Solution		Cu	smallest package
(133)	Iron standard		Fe	smallest package
(134)	Magnesium standard solution		Mg	smallest package
(135)	Manganese standard solution		Mn	smallest package

	<u>Material</u>	<u>Specifications</u>	<u>Chemical Formula</u>	<u>Quantity</u>
(136)	Sodium standard solution		Na	smallest package
(137)	Zinc standard solution		Zn	smallest package
(138)	Aluminium standard solution		Al	smallest package
(139)	Vanadium standard Solution		V	smallest package
(140)	Nickel standard solution		Ni	smallest package
(141)	Silicon standard solution		Si	smallest package
(142)	Lead standard solution		Pb	smallest package
(143)	Molybdenum standard solution		Mo	smallest package
(144)	Chromium standard solution		Cr	smallest package
(145)	Tin standards		Sn	smallest package

5. DESIGN CRITERIA

The main function of this laboratory maintenance of the power plant, as follows:

Provide testing capability to ensure that the plant equipment as a whole can be operated at high efficiency and capability.

Provide testing capability to ensure that all the plant equipment works properly to minimize forced outages and extend the life of the equipment.

The Central Laboratory shall be designed to perform including but not limited to the following operations:

Water analysis: This is the chemical and physical power plant cycle process, especially the analysis destruct and for decreases the unit performance, such as: conductivity, pH, silica, ammonium, hydrogen, iron, copper, phosphate, chemical oxygen demand, turbidity, etc.

6. DETAILED REQUIREMENTS

The equipment listed and specified on tables 1 to 2 are specially required under this Contract.

Bidder shall furnish an air conditioned building to house all the equipment. Additional requirements for this building are given in Part 4, 2.15 Structural Description and Criteria.

Bidder shall furnish the certificate for each of the precision equipment from an independent agency.

Bidder shall supply the required instrument and chemicals to perform the tests according to ASTM analyzing methods.

Bidder shall ensure that unnecessary duplication of equipment is avoided. It is intended that usually only one ASTM test specified is conducted at any one time. Consequently common equipment may be used for the different ASTM tests.

Any duplication of equipment furnished shall be clearly stated by the Bidder.

Bidder shall supply demineralizer water from Make up water tank to central laboratory and drainage water to the waste water treatment pond by sump pump.

Installation of the Central Laboratory shall be completed before the date when Steam Turbine operate and the plant shall be ready for operation at the time.

7. PERFORMANCE TESTING AND GUARANTEES

The Bidder shall conduct performance testing in accordance with Part 4, Subsection 2.6 Testing and Guarantees and shall guarantee the following system guarantees :

During commissioning of the equipment, Bidder will conduct all the ASTM tests specified in Clause 1.1 test to ascertain that the equipment is performing properly in every respect.

Any deficiencies in Bidder's equipment or material brought out during these tests shall be promptly remedied by Bidder at no expense to Owner.

Any modifications, changes, adjustments, or corrections which may be necessary in the opinion of the Owner or as may be requested by the Owner, shall be done by Bidder. This corrective work shall relate only to corrections of the defects based upon the original design of the equipment and shall be done without additional cost to the Owner.

Each part designated on the erection drawings by a mark number shall be clearly and legibly marked in a conspicuous place with the appropriate number if it is to be shipped separately.

Site Test

a. Running in Period Test

The equipment shall be functionally tested by the Bidder to the requirement of the Owner.

b. Functional Tests at Site

Volts, amperes and watts input shall be recorded for each applicable motor during all of the tests and compared with the values obtained during the shop tests.

c. Guarantee

Bidder shall guarantee the equipment performance at the design conditions.

8. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance/quality control shall conform to the requirements of Part 4, Subsection 2.7.

The quality assurance/Control documentation as specified in Part 4, Subsection 2.7 shall be included, but not limited to the following :

- a. Shop Test Reports
- b. Material Certifications
- c. NDE Reports
- d. All other documentation required by applicable codes and standards.

9. INSTRUMENTATION AND CONTROLS

Instrumentation and controls provided as specified and/or required by the Bidder to ensure the proper function and safety of the operation shall be in accordance with Technical Specification I-001 and I-002 where ever applicable.

10. ELECTRICAL SPECIFICATION

The laboratory building electrical installation shall be accordance with Technical Specifications E-001, E-230, E-232, E-241, E-270, and E-271.

(Last page of M-422)